Module 6

SPI — UVM+SV Verification

Learning objectives

- SPI UVM agent: Transaction (SpiTransaction), sequence (SpiSequence), driver (SpiDriver), monitor (S…
- Interface: spi_master_if (clk, rst_n, clk_div_tick, start, data_in, sclk, mosi, cs_n, busy, done) c…
- Mode 0 monitoring: SCLK idle low; capture on rising edge; monitor samples MOSI on each rising SCLK,…

Prerequisites

- Module 1 (spec → RTL), Module 2 (UVM+SV, uvm_smoke), Module 5 (SPI protocol + RTL + basic TB).
- Verilator, Make, C++ compiler, UVM (UVM_HOME or vendored UVM).

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

```
flowchart LR
T1["SPI UVM Agent"]
T2["Mode 0 Timing"]
T1 --> T2
T3["Toolchain"]
T2 --> T3
```

Extend SPI verification to **UVM+SV** — SPI agent (transaction, sequence, driver, monitor, scoreboard); run on Verilator.

Overview

- Module 6 builds on Module 5 (SPI protocol + RTL + basic testbench):

Syllabus topics

1. SPI UVM Agent

- Transaction (SpiTransaction): byte to send (data); monitor fills observed_mosi (reconstructed from…
- Sequence (SpiSequence): produces directed transactions (0x00, 0x01, 0x55, 0xAA, 0xFF).
- Driver (SpiDriver): gets transaction, drives start and data_in for one cycle, waits for done.
- Monitor (SpiMonitor): watches cs_n; when cs_n goes low, samples MOSI on each rising edge of SCLK (m…
- Scoreboard (SpiScoreboard): compares expected (from test) vs observed_mosi (from monitor).

2. Mode 0 Timing

- SPI mode 0: SCLK idles low; data captured on rising edge; data changed on falling edge.
- Monitor: Waits for cs_n low, then on 8 rising edges of SCLK captures MOSI into bits [7:0] (MSB firs…

3. Toolchain

- Same as Module 2: Verilator with -sv, --timing, --trace, UVM include paths, make SIM=verilator TEST…

Hands-on examples

Module 6 self-check

```
$ ./scripts/module6.sh --check
```

Terminal: module6_check

```
□[0;36m========================================□[0m
□[0;36mModule 6: Demo commands (SPI UVM example)□[0m
□[0;36m========================================□[0m

From repo root:

# 1) Environment sanity:
verilator --version
make --version
echo "$UVM_HOME"

# 2) Run the SPI UVM example (inside the example directory):
cd module6/examples/spi_uvm
make SIM=verilator TEST=test_spi_uvm

# 3) Inspect logs and waveforms:
ls obj_dir
# If a VCD is produced:
ls *.vcd

□[1;33m[INFO]□[0m See module6/EXAMPLES.md and module6/examples/spi_uvm/README.md for more details.
```

Example 1: SPI UVM

- UVM phases and reporting (DRIVER, MONITOR, SCOREBOARD).
- Scoreboard checks (expected byte from sequence vs observed byte on MOSI).
- Final scoreboard summary (matches and mismatches).

Demo: SPI UVM

```
$ ./scripts/module6.sh --run
```

Terminal: ex01_spi_uvm

```
## Success

You should see UVM phases, DRIVER/MONITOR/SCOREBOARD messages, and a scoreboard summary with 5 match
```

Practice & assessment

Exercises

- Run spi_uvm
- Trace one transaction
- Change the sequence
- Optional: Compare to UART UVM

Assessment checklist

- Can describe the SPI UVM agent and how it maps to SPI (one byte, mode 0 monitoring).
- Can run spi_uvm and interpret UVM output.
- Ready for Module 7: I²C protocol + RTL + basic testbench.

Summary & next steps

- Extend SPI verification to **UVM+SV** — SPI agent
  (transaction, sequence, driver, monitor, scoreboa…
- Complete module6/CHECKLIST.md
- Run ./scripts/module6.sh --check
- Next: I²C